7 January 1970
SUPERSEDING
MIL-W-43061A(GL)
26 February 1965

MILITARY SPECIFICATION

WAX, MACHINE STITCHING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.-This specification covers a stitching wax of one type and grade (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-290 - Packaging, Packing and Marking of Petroleum and Related Products.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS, (ASTM) DESIGNATION

E28-67 - Method of Test for Softening Point by Ring and Ball Apparatus.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, ATTN: Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.-The material shall be made of plasticized rosin so blended as to meet the requirements specified below. The material shall function properly on a standard shoe repair machine equipped with a device for melting wax.

3.1.1 Detrimental effects.-The wax used as a thread dressing shall not be detrimental to the thread or leather.

* 3.2 Consistency.-The consistency of the wax shall be such that its softening point shall be 115° to 130°F. when tested as specified in 4.2.4.

* 3.3 Color.-The wax shall be black, white, brown or golden in color as specified in the contract or order (see 6.2).

* 3.4 Odor.-The wax shall not have a rancid odor.

3.5 Performance.-The molten wax shall adhere to the thread and when cooled and used in the shoe sewing operation, shall hold to the leather.

* 3.6 Workmanship.-The finished wax shall be clean and homogeneous in appearance.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

\# 4.1.1 Certificate of compliance.-Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection.-Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Inspection of materials and components.-In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.2.2 Inspection of the end item.-

* 4.2.2.1 Examination of the end item.-The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth in 4.2.3. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of one pound cakes of wax for the examinations in 4.2.2.1.1 and 4.2.2.1.2 and in units of shipping containers for the examination in 4.2.2.1.3.

* 4.2.2.1.1 Examination of the end item for defects in color, workmanship, and odor.-The sample unit for this examination shall be a one pound cake of wax.

Examine	Defects
Color	Not as specified.
Odor	Rancid.
Workmanship	Not homogeneous, not clean, contains dirt.

* 4.2.2.1.2 Examination of the end item for defects in net weight.-The sample unit for this examination shall be a one pound cake of stitching wax. The lot shall be unacceptable if the average net weight is less than indicated or specified.

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* 4.2.2.1.3 Examination of preparation for delivery.-An examination shall be made to determine that packaging, packing and marking requirements comply with the requirements of Section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers fully prepared for delivery shall be examined for closure defects. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defects
Markings	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence or method of application. Not in accordance with contract requirements (see 5.3).
Materials	Any nonconforming component; component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as: incomplete closure of case liners, container flaps, loose or inadequate sealing, strapping or stapling.
	Bulged or distorted container.
Contents	Wax slabs not packed as specified (see 5.2).
Weight	Net weight of contents below minimum requirement.

4.2.3 Inspection levels and acceptable quality levels (AQL's) for examinations.-The inspection levels for determining the sample size and the acceptable quality levels (AQL's) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	S-2	2.5
4.2.2.1.2	S-2	N.A.
4.2.2.1.3	S-2	4.0

* 4.2.4 Testing of the end item.-A composite sample of the end item shall be tested for the characteristics listed in table 1 for each lot. For purposes of sampling, the lot shall be expressed in pounds of wax. The sample unit for testing shall be a 2-pound composite obtained by combining equal portions from samples selected at random throughout the lot. The sample size shall be one composite sample. The portions selected for the composite sample shall be melted and the mixture subjected to continuous stirring while being cooled to ambient temperature. The composite shall be placed in a clean, dry, sealed container. Care shall be exercised to prevent contamination or alteration of the composite during the sampling, compositing, storage and testing. All test reports shall contain the individual values utilized in expressing the final

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result. The lot shall be unacceptable if the composite fails to meet any test requirement specified.

NOTE: The portions selected for the composite sample shall be drawn from the lot at the frequency indicated below:

Lot size (pounds)	Sample frequency
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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INSTRUCTIONS FOR TESTING OF THE END ITEM

TABLE I

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per sample unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Material	3.1	1/							
Deteriorating effects	3.1.1	1/							
Consistency	3.2	4.3.1			Average of 2		1.0°F.		
Performance	3.5	1/							

1/ Supplier shall furnish a certificate of compliance covering this requirement.

AMXRE Form 598
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4.3 Test procedures.-

4.3.1 Consistency.-The softening point of the wax shall be tested in accordance with ASTM Designation E28-67.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-Packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A.-Each 1-pound cake of wax shall be completely wrapped or placed in a snug-fitting bag. The bag shall be closed in such a manner as to retain the wax during transit. The wrap or bag shall be compatible with the wax to minimize sticking to the wax.

* 5.1.2 Level C.-Wax shall be packaged to afford adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing.-Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A.-Fifty 1-pound cakes, packaged as specified in 5.1, shall be packed in the most compact manner in a snug-fitting fiberboard shipping container conforming to grade V2s, style RSC-L of PPP-B-636. The inside of each shipping container shall be fitted with a full-height taped box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-F-320. Shipping containers shall be closed, waterproofed by means of tape, and reinforced in accordance with the appendix of PPP-B-636.

* 5.2.2 Level B.-Fifty 1-pound cakes, packaged as specified in 5.1, shall be packed in the most compact manner in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be fitted with a full-height taped box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-F-320. Shipping containers shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C.-The wax, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Classification Rules or National Motor Freight Classification Rules, as applicable.

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5.3 Marking.-In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-290.

5.3.1 Precautionary marking.-In addition to marking as specified in 5.3, each shipping container shall bear the following precautionary marking on one end and one side in letters not less than 1 inch high:

"STORE IN COOL PLACE"

6. NOTES

6.1 Intended use.-The wax covered by this specification is intended to cover the requirements for machine stitching in the manufacture and repair of shoes.

6.2 Ordering data.-Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Color required (see 3.3).

(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Basis of purchase.-A container shall have fifty cakes of wax, approximately one pound each. Payment shall be based upon net weight.

6.4 The outside margins of this document have been marked to indicate where changes from the previous issue have been made. An asterisk (*) indicates a substantive change in the paragraph text. A double asterisk (**) indicates deletion of a paragraph bearing the same number. The number sign (#) indicates the addition of a completely new paragraph. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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